



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

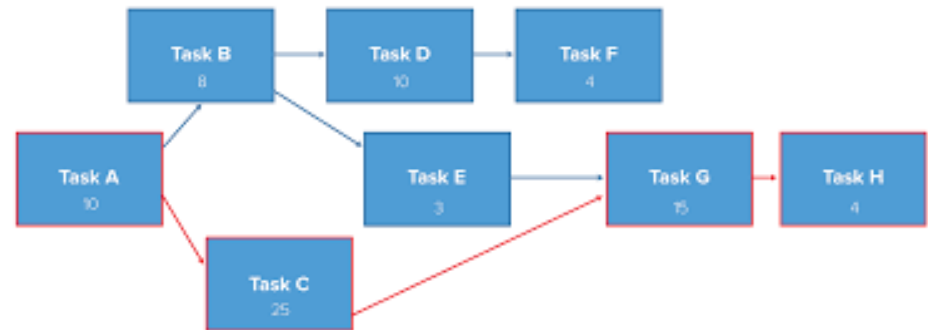
Prof Masoumeh Kazemi



Agenda

- Project Management-Chapter 17

Project Management

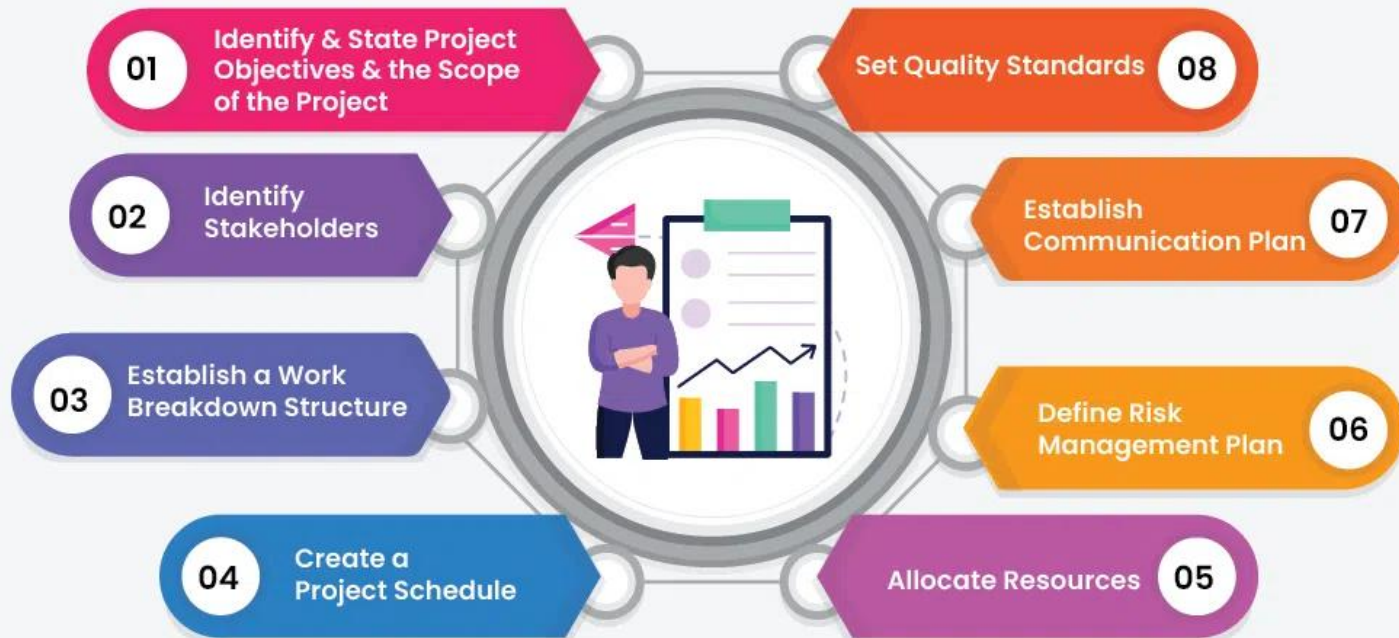


- A project consists of a set of activities
- The activities may start or stop independent of other activities
- Activities have precedence relationships and must be performed in the proper order
- Project Management: planning and organization of resources to move a specific project towards completion at **minimum cost**
- Applications in construction and other engineering projects

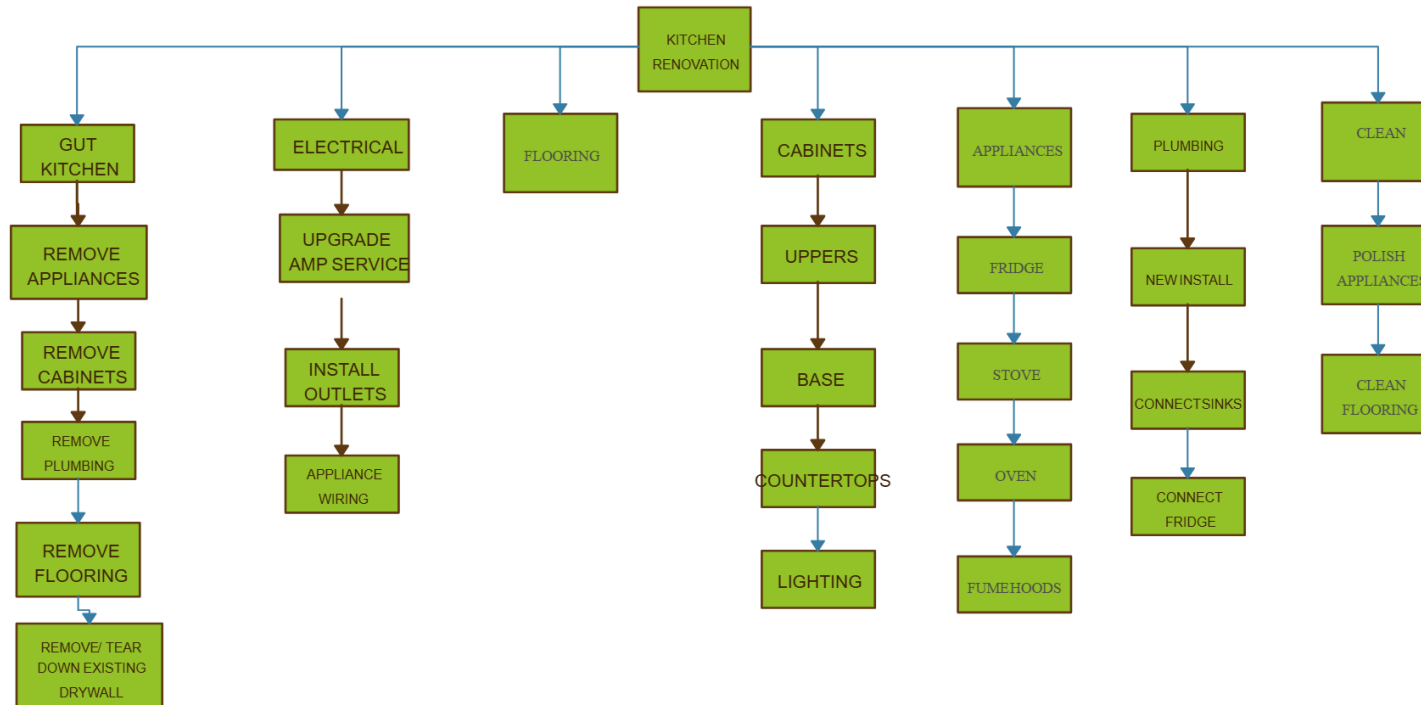
Project Management



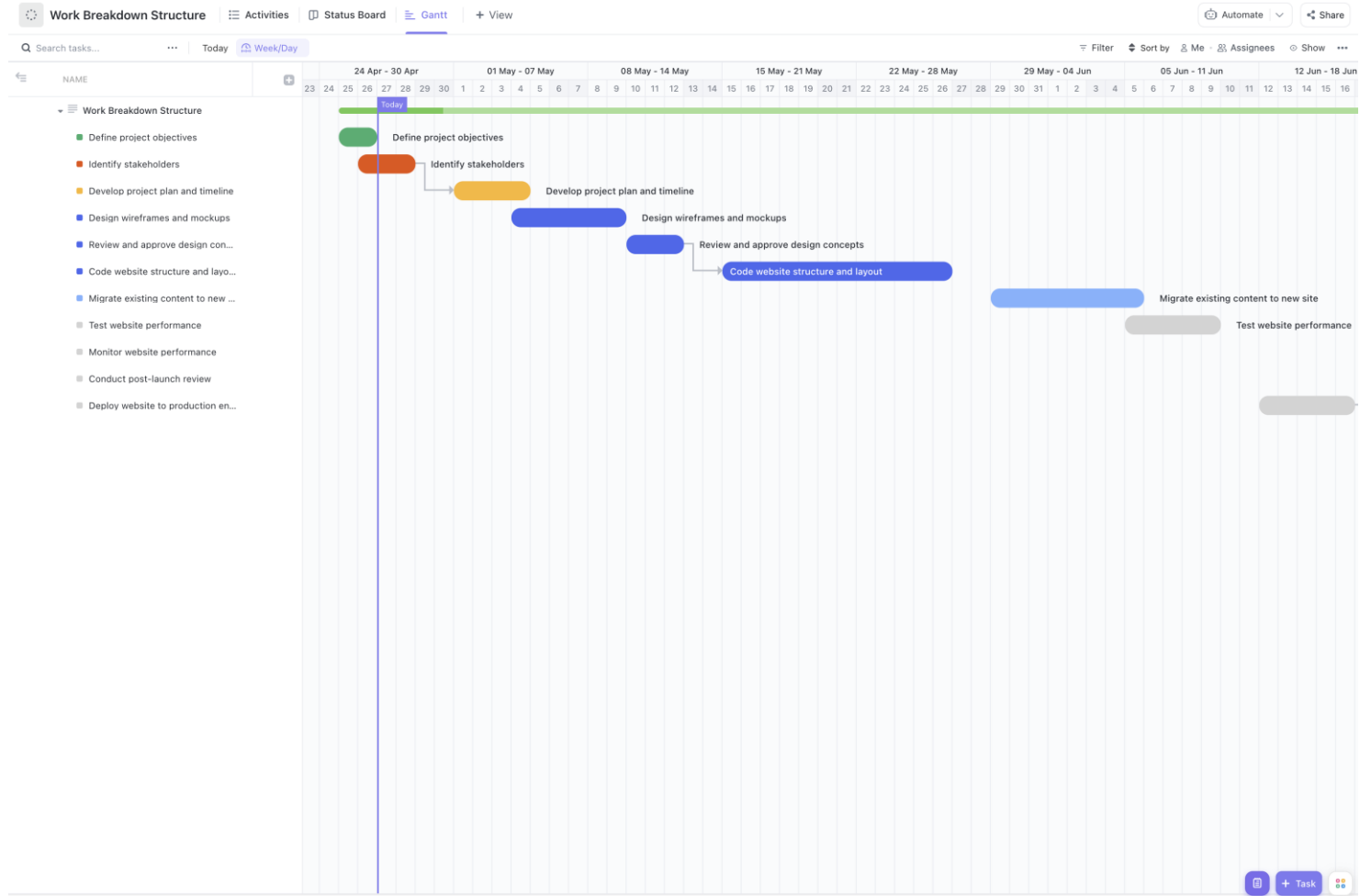
How to Create a Project Management Plan?



Example of Work Breakdown Structure (WBS)

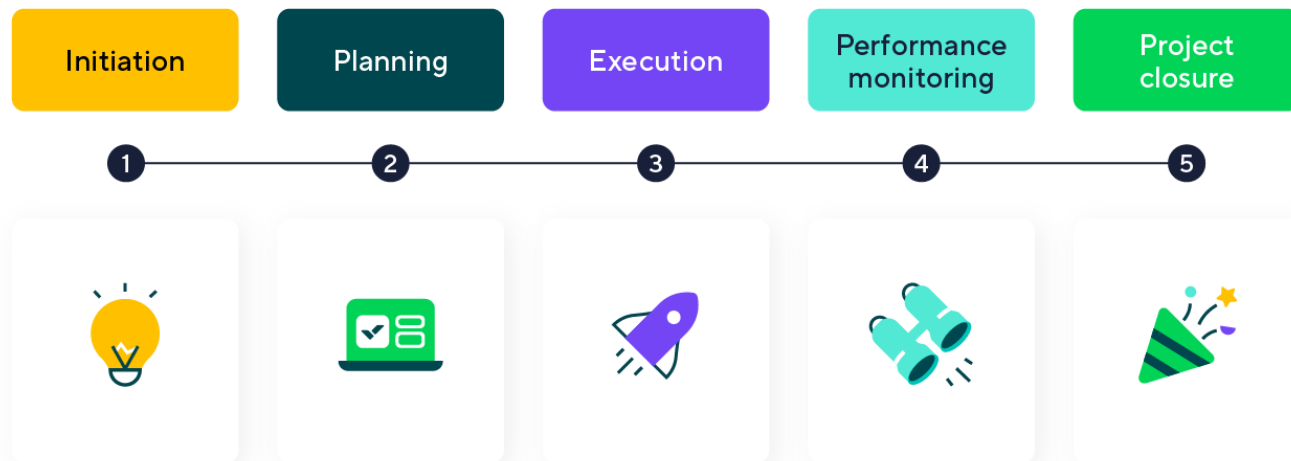


Example of Project schedule



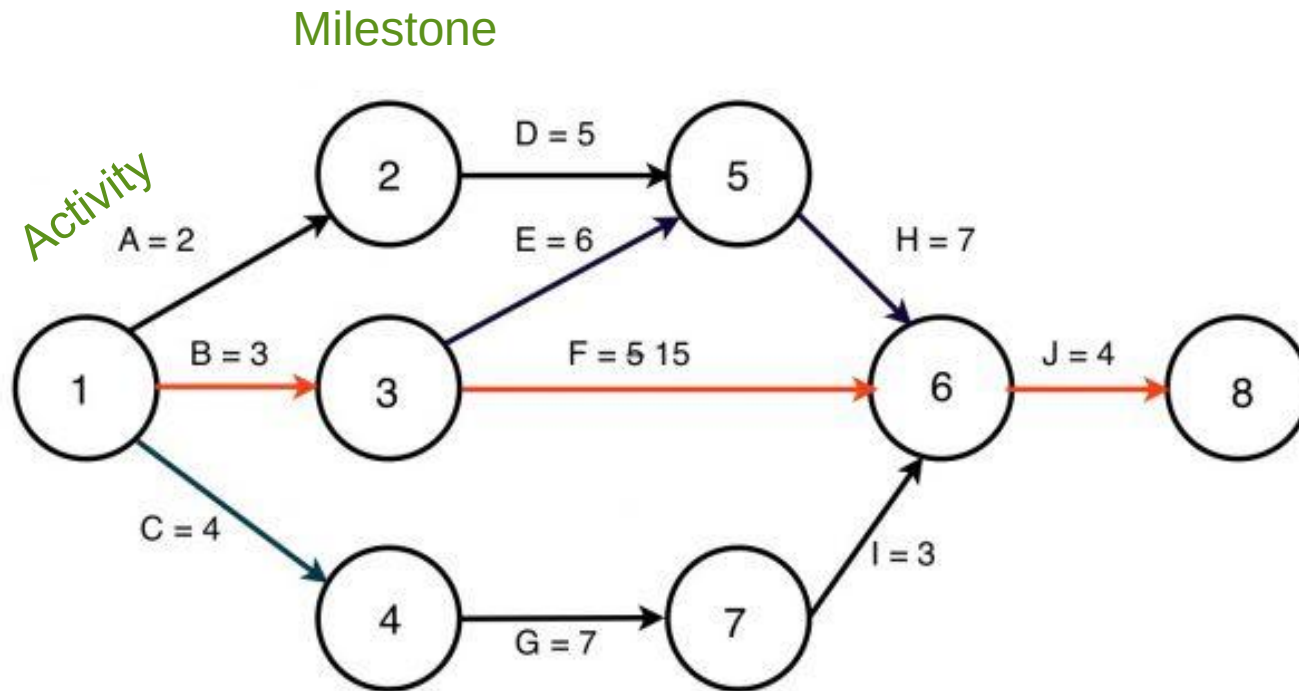
Project Management

The 5 Phases of a Project Management Plan



Project Management

- Example



Project Management

- Critical Path Method (CPM)
 - Use deterministic information
 - To identify critical path in completing a project
 - The longest sequence of tasks that determines the minimum project duration
- Program Evaluation and Review Technique (PERT)
 - Data are presented with probabilities
 - Results are also probabilistic

A CPM example with deterministic information

- *A*, *B* and *C* are concurrent and begin at the start of the project
- *D* and *E* are concurrent and can begin upon the completion of *A*
- *F* may begin upon the completion of *B*
- *G* starts after *C* is finished
- *H* follows *D*
- *I* can start after *E* and *F* are completed
- The project is complete after *G*, *H* and *I* are complete

- Additionally:

A: 8 days	F: 9 days
B: 20 days	G: 10 days
C: 33 days	H: 8 days
D: 18 days	I: 4 days
E: 20 days	

The CPM network of the example

A: 8 days

B: 20 days

C: 33 days

D: 18 days

E: 20 days

F: 9 days

G: 10 days

H: 8 days

I: 4 days

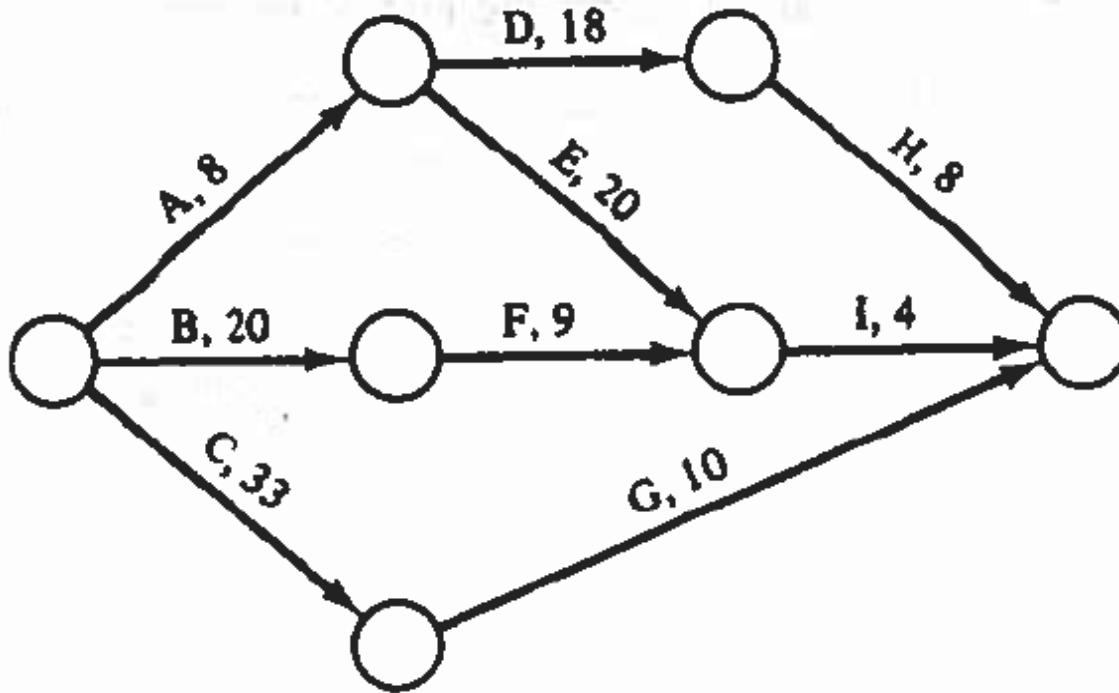


Fig. 17.2. CPM network.

The forward pass

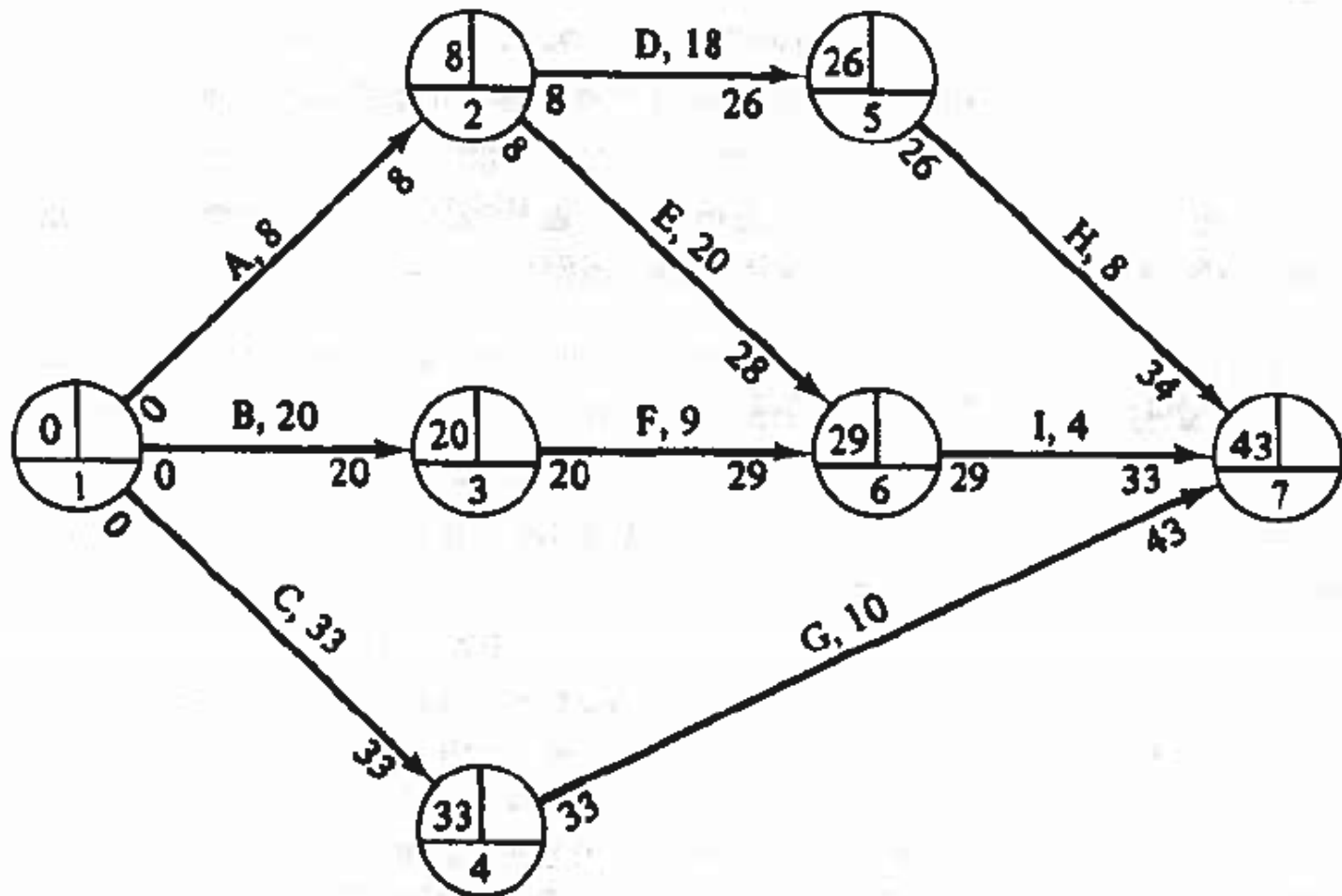


Fig. 17.4. Forward pass calculations.

Forward and backward pass calculations

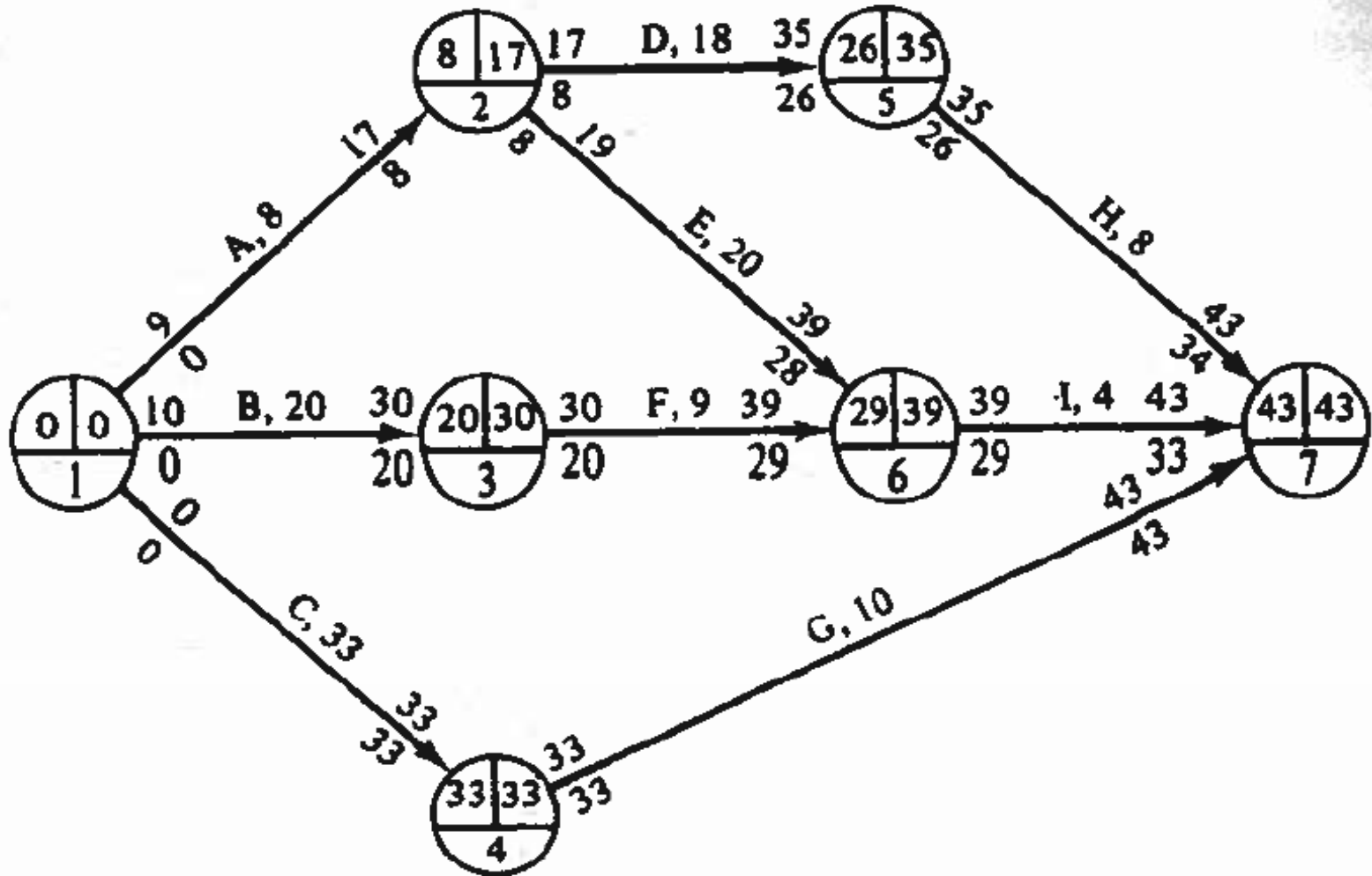


Fig. 17.6. Forward and backward pass calculations.

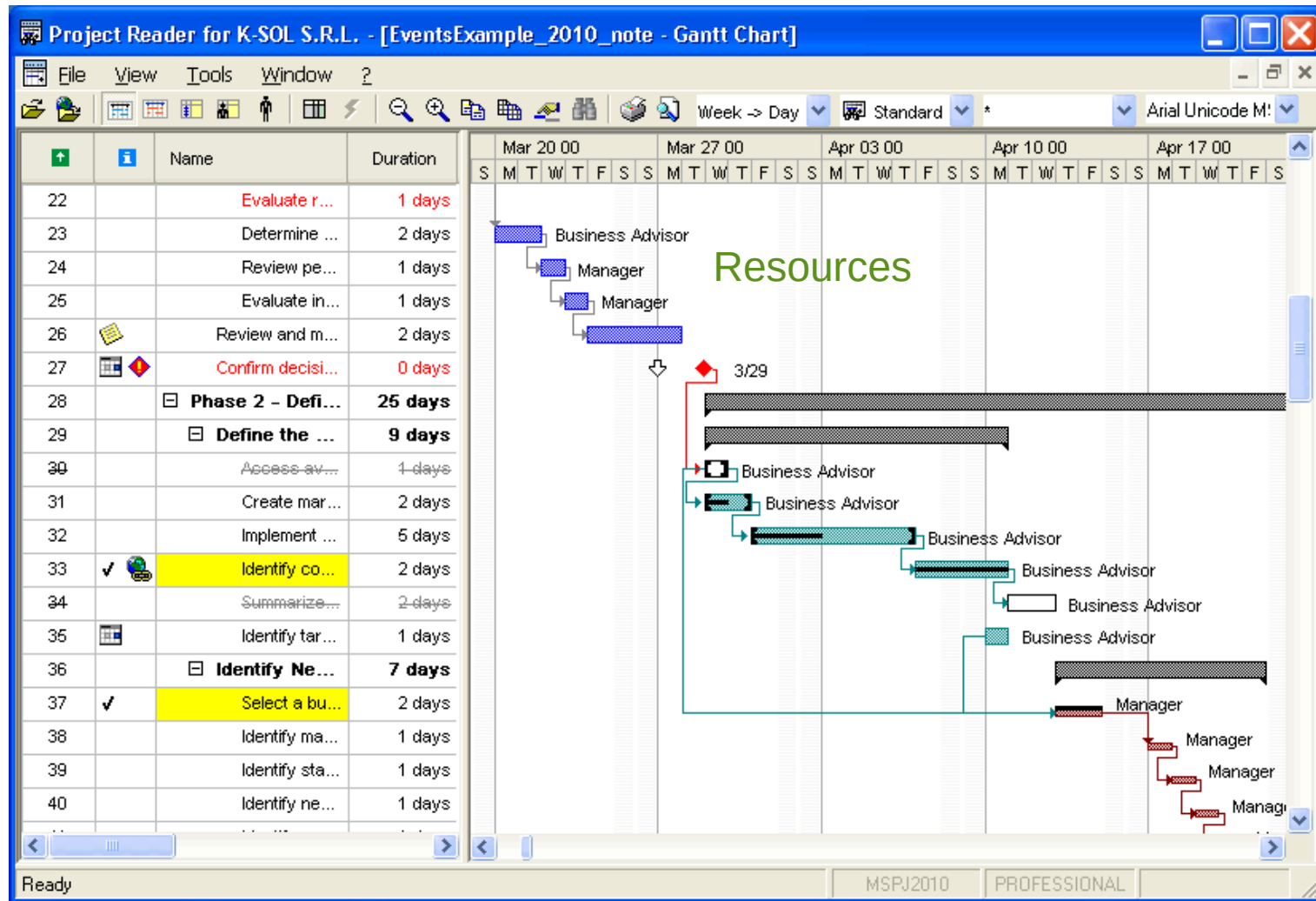
Forward and backward pass notations



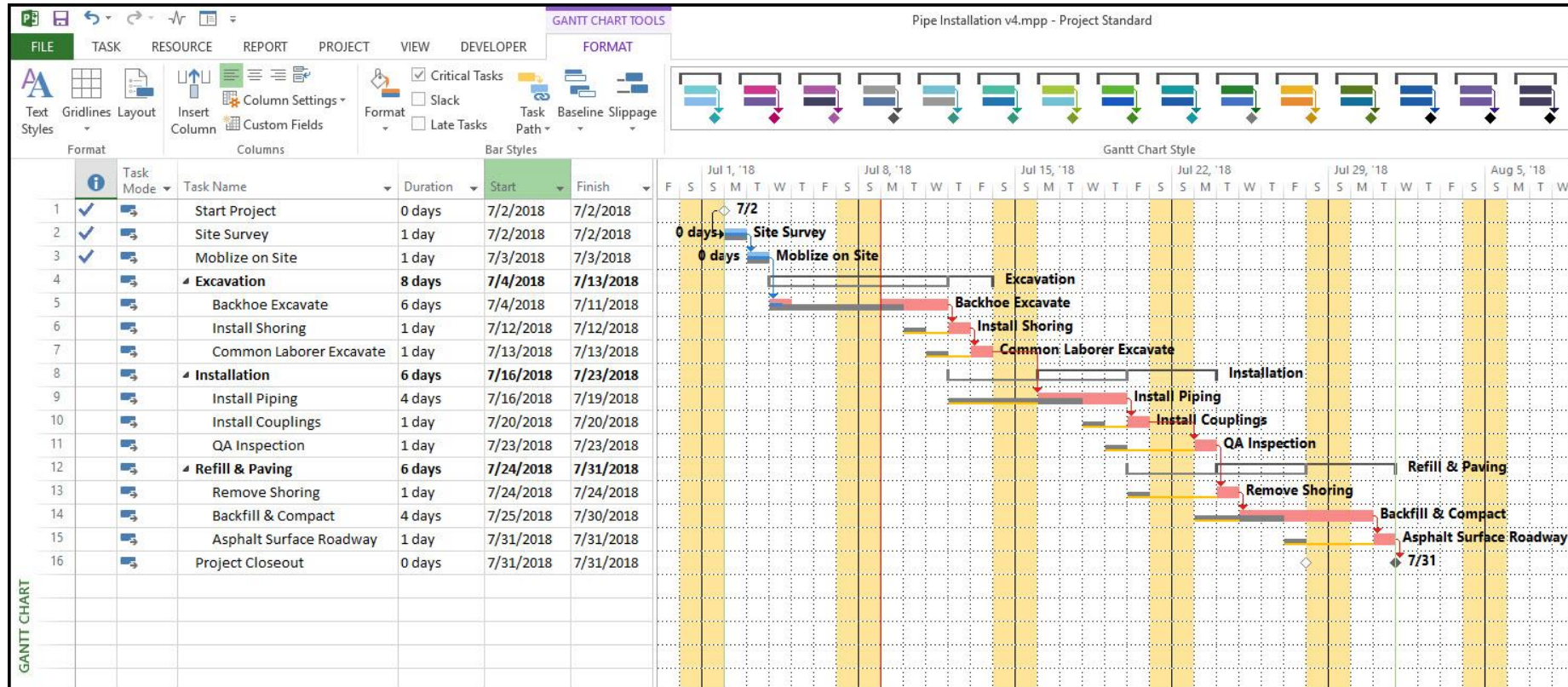
Fig. 17.5. Forward and backward pass notation.

- t = single estimate of mean activity duration
- T_E = earliest event occurrence time
- T_L = latest allowable event occurrence time
- ES = earliest activity start time
- EF = earliest activity finish time
- LS = latest allowable activity start time
- LF = latest allowable activity finish time
- S = total activity slack

Project Management software



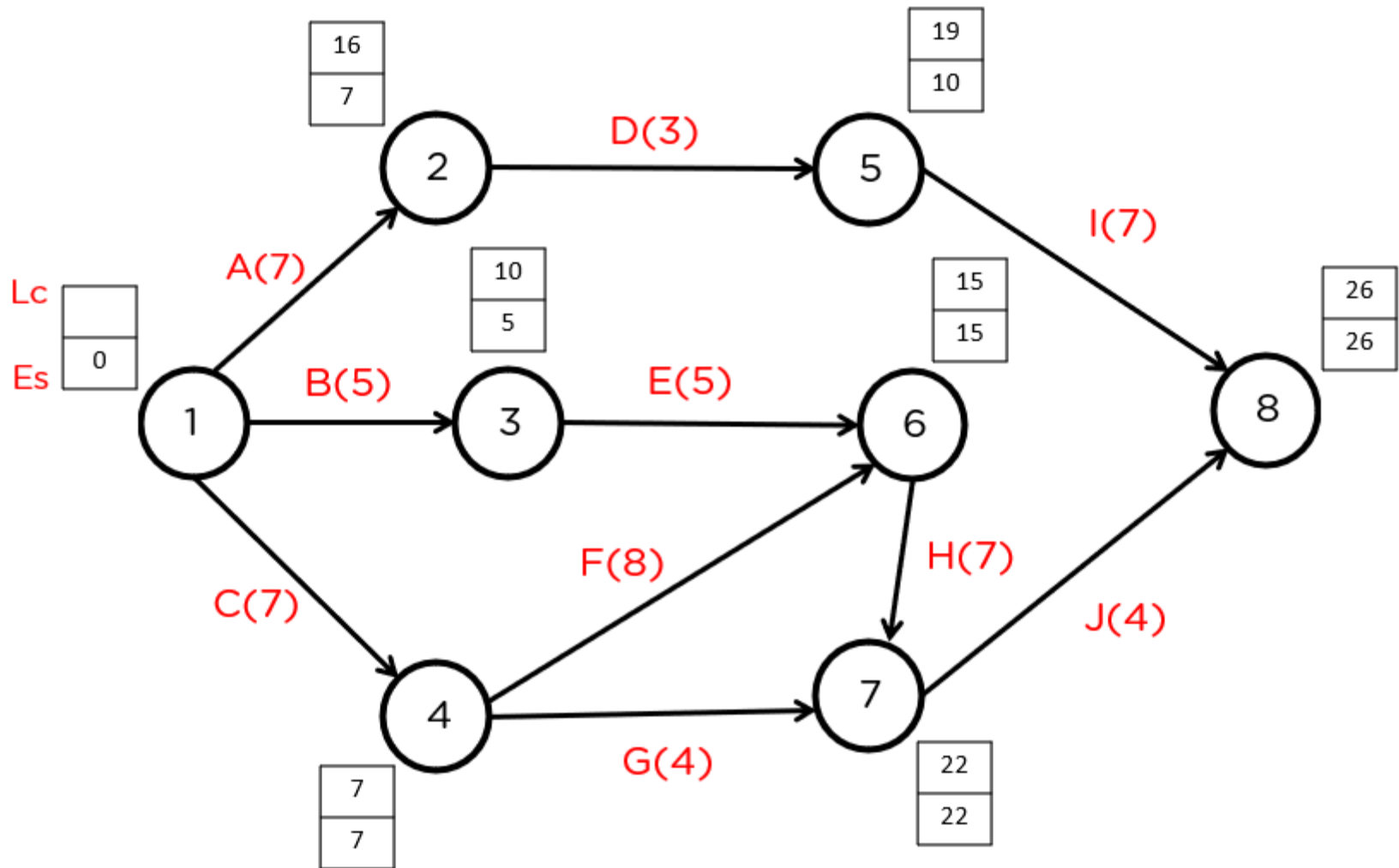
Project Management software



PERT

- Program Evaluation and Review Technique
 - A project management tool used to schedule, organize, and coordinate tasks for projects with uncertain timelines
 - Uses a network diagram (PERT chart)
 - Accounts for uncertainty by using 3 time estimates for each task:
 - Optimistic time
 - Most likely time
 - Pessimistic time
 - For each activity, the expected duration is calculated based on the above estimations, using a weighted average
 - The variance of each activity can be also calculated
 - The critical path is determined based on the expected duration of events
 - The probability of finishing the project by a certain date can be estimated accordingly

PERT Chart



Other related issues

- Adding resources to reduce activity time
- New critical path (bottleneck) will emerge
- Use linear programming to optimize
- Use simulation if probabilistic
- Resource constrained project scheduling
- Monitoring and enforcing the deadlines